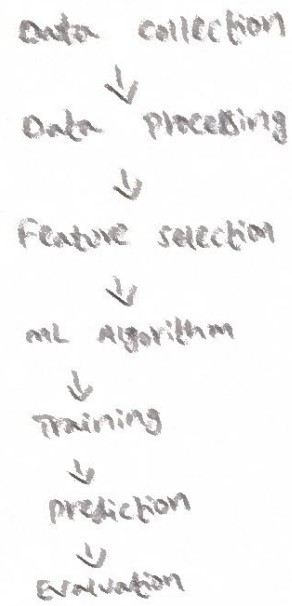


Cod - Assessment Tool - 02 - Concept Mapping

1A) A machine learning system consists of the following major blocks



Data collection:- collect data from sensors, data bases or files

Pre-processing:- Remove missing, duplicate and using data

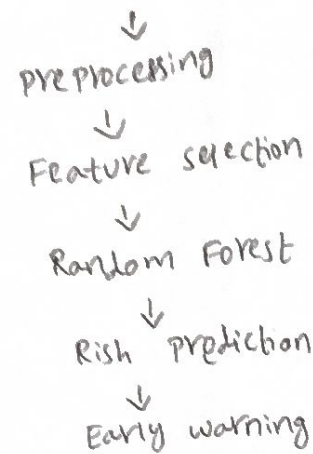
Feature selection:- select important features

Training:- The algorithm learns patterns from training data

Prediction:- The trained model predicts outputs for new data

2A) ML system for disaster management:-

Historical Real time data



→ Historical rainfall, water level, temp and data collected

→ Missing and noisy values are removed

→ Important feature are selected

→ Data is divided into training, validation and testing sets.

→ If high risk is predicted, an alert is sent to authorities and the public.

3A) Positive examples:-

1. overcast, not, high, weak
2. Rainy, mild, high weak
3. Rainy, cool, Normal weak
4. overcast, mild, Normal, strong

Hypothesis updates:-

$[S - 0 = \langle \text{angle} \setminus \text{empty set}, \setminus \text{empty set}, \setminus \text{empty set} \setminus \text{empty set} \setminus \text{angle} \rangle]$

After ex 3:-

$[S - 1 = \langle \text{angle overcast, not, high, weak} \setminus \text{angle} \rangle]$

After ex 4:-

$[S - 2 = \langle \text{angle} ?, ? \text{high, weak} \setminus \text{angle} \rangle]$

After ex 5:-

$[S - 3 = \langle \text{angle} ?, ?, ? \text{weak} \setminus \text{angle} \rangle]$

After ex 7:-

$[S - 4 = \langle \text{angle} ??? \setminus \text{angle} \rangle]$

Final Answer:-

$[boxed \{S = \langle \text{angle} ??? \setminus \text{angle} \rangle\}]$

10A)

Patient Data



Preprocessing



Feature selection



Train/Test split



Random Forest / Logistic Regression



Disease Prediction



Evaluation

→ Patient data such as age, blood pressure, glucose & life style are collected

→ missing values & duplicate records are handled

→ categorical data is encoded

→ Important features are selected

→ Logistic Regression is suitable for binary Disease Prediction

→ Random forest can capture non-linear relationships

→ The dataset is divided into training and testing sets.

11A) Data collection



Pre processing



Feature Engineering



model training



Evaluation



Deployment



Monitoring

→ Data collection:- obtains raw data

→ preprocessing:- cleans and transforms the data

→ Feature Engineering:- creates useful features

→ Training:- ML algorithm learns patterns

→ Evaluation:- measures model performance

→ Deployment:- uses the model in a real Application

→ monitoring:- continuously checks model performance

Example:-

[

Training \ Accuracy = 98%

]

[

Testing \ Accuracy = 70%

]